

Web appendices for “Prior Selection for Vector Autoregressions”

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A The Marginal Likelihood of BVARs with Conjugate Priors

This appendix derives an analytical expression for the ML of BVARs with conjugate priors (possibly implemented using dummy observations), and proves the fit-complexity trade-off result that we have stated in (2.5) in the main text.

A.1 Analytical derivation of the ML

Consider the VAR model of section 1

$$\begin{aligned} y_t &= C + B_1 y_{t-1} + \dots + B_p y_{t-p} + \varepsilon_t, \quad t = 1, \dots, T \\ \varepsilon_t &\sim N(0, \Sigma), \end{aligned}$$

and rewrite it as

$$\begin{aligned} Y &= X\beta + \epsilon \\ \epsilon &\sim N(0, \Sigma \otimes I_{T-p}), \end{aligned}$$

where $y \equiv [y_{p+1}, \dots, y_T]'$, $Y \equiv \text{vec}(y)$, $x_t \equiv [1, y'_{t-1}, \dots, y'_{t-p}]'$, $X_t \equiv I_n \otimes x'_t$, $x \equiv [x_{p+1}, \dots, x_T]'$, $X \equiv I_n \otimes x$, $\varepsilon \equiv [\varepsilon_{p+1}, \dots, \varepsilon_T]'$, $\epsilon \equiv \text{vec}(\varepsilon)$, $B \equiv [C, B_1, \dots, B_p]'$ and $\beta \equiv \text{vec}(B)$. Finally, define the number of regressors for each equation by $k \equiv np + 1$.

As in section 2, the prior on (β, Σ) is given by the following Normal-Inverse-Wishart

distribution¹

$$\begin{aligned}\Sigma &\sim IW(\Psi, d) \\ \beta|\Sigma &\sim N(b, \Sigma \otimes \Omega),\end{aligned}$$

where, for simplicity, we are not explicitly conditioning on the hyperparameters b , Ω , Ψ and d .

The unnormalized posterior of (β, Σ) can be obtained by multiplying the prior density by the likelihood function. If we condition on the initial p observations of the sample, which is a standard assumption, we obtain:

$$\begin{aligned}p(\beta, \Sigma|Y) &= \left(\frac{1}{2\pi}\right)^{\frac{n(T-p+k)}{2}} |\Sigma|^{-\frac{T-p+k+n+d+1}{2}} |\Omega|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} \frac{e^{-\frac{1}{2}\text{tr}(\Psi\Sigma^{-1})}}{2^{\frac{nd}{2}} \cdot \Gamma_n\left(\frac{d}{2}\right)} \\ &\quad e^{-\frac{1}{2}\left[\begin{aligned} &(Y - X\beta)'(\Sigma \otimes I_T)^{-1}(Y - X\beta) + \\ &+ (\beta - b)'(\Sigma \otimes \Omega)^{-1}(\beta - b) \end{aligned}\right]}. \end{aligned} \quad (\text{A.1})$$

Tedious algebraic manipulations of (A.1) yield the expression

$$\begin{aligned}p(\beta, \Sigma|Y) &= \left(\frac{1}{2\pi}\right)^{\frac{n(T-p+k)}{2}} |\Sigma|^{-\frac{T-p+k+n+d+1}{2}} |\Omega|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} \frac{e^{-\frac{1}{2}\text{tr}(\Psi\Sigma^{-1})}}{2^{\frac{nd}{2}} \cdot \Gamma_n\left(\frac{d}{2}\right)} \\ &\quad e^{-\frac{1}{2}\left[\begin{aligned} &(\beta - \hat{\beta})' \left[X'(\Sigma \otimes I_T)^{-1}X + (\Sigma \otimes \Omega)^{-1} \right] (\beta - \hat{\beta}) + \\ &+ (\hat{\beta} - b)'(\Sigma \otimes \Omega)^{-1}(\hat{\beta} - b) + \hat{\epsilon}'(\Sigma \otimes I_T)^{-1}\hat{\epsilon} \end{aligned}\right]} \end{aligned} \quad (\text{A.2})$$

where $\hat{B} \equiv (x'x + \Omega^{-1})^{-1}(x'y + \Omega^{-1}b)$, $\hat{\beta} \equiv \text{vec}(\hat{B})$, $\hat{\epsilon} \equiv y - x\hat{B}$, $\hat{\epsilon} \equiv \text{vec}(\hat{\epsilon})$, and b is a $k \times n$ matrix obtained by reshaping the vector b in such a way that each column corresponds to the prior mean of the coefficients of each equation (i.e. $b \equiv \text{vec}(b)$). It can then be shown that (A.2) is the kernel of the following Normal-Inverse-Wishart posterior distribution:

$$\Sigma|Y \sim IW\left(\Psi + \hat{\epsilon}'\hat{\epsilon} + (\hat{B} - b)' \Omega^{-1}(\hat{B} - b), T - p + d\right) \quad (\text{A.3})$$

$$\beta|\Sigma, Y \sim N\left(\hat{\beta}, \Sigma \otimes (x'x + \Omega^{-1})^{-1}\right). \quad (\text{A.4})$$

The ML is the integral of the unnormalized posterior:

$$p(Y) = \int \int p(Y|\beta, \Sigma) \cdot p(\beta|\Sigma) \cdot p(\Sigma) d\beta d\Sigma. \quad (\text{A.5})$$

¹We are using the following parameterization of the Inverse Wishart density: $p(\Sigma|\Psi, d) = \frac{|\Psi|^{\frac{d}{2}} \cdot |\Sigma|^{-\frac{n+d+1}{2}} \cdot e^{-\frac{1}{2}\text{tr}(\Psi\Sigma^{-1})}}{2^{\frac{nd}{2}} \cdot \Gamma_n\left(\frac{d}{2}\right)}$.

Let's start with the integral with respect to β . Substituting (A.2) into (A.5) we obtain

$$p(Y, \Sigma) = \int \left[\begin{array}{c} \left(\frac{1}{2\pi} \right)^{\frac{n(T-p+k)}{2}} |\Sigma|^{-\frac{T-p+k+n+d+1}{2}} |\Omega|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} \frac{e^{-\frac{1}{2}tr(\Psi\Sigma^{-1})}}{2^{\frac{nd}{2}} \cdot \Gamma_n\left(\frac{d}{2}\right)} \\ -\frac{1}{2} \left[\begin{array}{c} (\beta - \hat{\beta})' \left[X'(\Sigma \otimes I_{T-p})^{-1} X + (\Sigma \otimes \Omega)^{-1} \right] (\beta - \hat{\beta}) + \\ + (\hat{\beta} - b)' (\Sigma \otimes \Omega)^{-1} (\hat{\beta} - b) + \hat{\epsilon}' (\Sigma \otimes I_{T-p})^{-1} \hat{\epsilon} \end{array} \right] \end{array} \right] d\beta,$$

which can be solved by “completing the squares,” yielding

$$p(Y, \Sigma) = \left(\frac{1}{2\pi} \right)^{\frac{n(T-p)}{2}} |\Sigma|^{-\frac{T-p+n+d+1}{2}} |\Omega|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} \frac{e^{-\frac{1}{2}tr(\Psi\Sigma^{-1})}}{2^{\frac{nd}{2}} \cdot \Gamma_n\left(\frac{d}{2}\right)} \cdot e^{-\frac{1}{2}[(\hat{\beta}-b)'(\Sigma \otimes \Omega)^{-1}(\hat{\beta}-b) + \hat{\epsilon}'(\Sigma \otimes I_{T-p})^{-1}\hat{\epsilon}]} \cdot |x'x + \Omega^{-1}|^{-\frac{n}{2}}.$$

We are now ready to take the integral with respect to Σ :

$$p(Y) = \left(\frac{1}{2\pi} \right)^{\frac{n(T-p)}{2}} |\Omega|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} \frac{1}{2^{\frac{nd}{2}} \cdot \Gamma_n\left(\frac{d}{2}\right)} |x'x + \Omega^{-1}|^{-\frac{n}{2}} \int \left[\begin{array}{c} |\Sigma|^{-\frac{T-p+n+d+1}{2}} e^{-\frac{1}{2}tr(\Psi\Sigma^{-1})} \\ -\frac{1}{2} \underbrace{\left[(\hat{\beta} - b)' (\Sigma \otimes \Omega)^{-1} (\hat{\beta} - b) + \hat{\epsilon}' (\Sigma \otimes I_{T-p})^{-1} \hat{\epsilon} \right]}_P \end{array} \right] d\Sigma. \quad (\text{A.6})$$

The expression for P can be simplified by using the following property of the *vec* operator:

$$vec(A)' (D \otimes B) vec(C) = tr(A'BCD').$$

This yields

$$P = tr \left[\hat{\epsilon}' \hat{\epsilon} \Sigma^{-1} + (\hat{B} - b)' \Omega^{-1} (\hat{B} - b) \Sigma^{-1} \right]. \quad (\text{A.7})$$

We can now solve the integral by substituting (A.7) into (A.6), and multiplying and dividing the expression inside the integral by the constant term necessary to obtain the density of an Inverse-Wishart. These operations result in the following closed-form solution for the ML:

$$p(Y) = \left(\frac{1}{\pi} \right)^{\frac{n(T-p)}{2}} \frac{\Gamma_n\left(\frac{T-p+d}{2}\right)}{\Gamma_n\left(\frac{d}{2}\right)} \cdot |\Omega|^{-\frac{n}{2}} \cdot |\Psi|^{\frac{d}{2}} \cdot |x'x + \Omega^{-1}|^{-\frac{n}{2}} \cdot \left| \Psi + \hat{\epsilon}' \hat{\epsilon} + (\hat{B} - b)' \Omega^{-1} (\hat{B} - b) \right|^{-\frac{T-p+d}{2}}. \quad (\text{A.8})$$

A.2 Numerical issues

For large systems (in our case, for the medium- and the large-scale models), (A.8) is numerically unstable and it is convenient to replace it with the equivalent expression

$$p(Y) = \left(\frac{1}{\pi}\right)^{\frac{n(T-p)}{2}} \frac{\Gamma_n\left(\frac{T-p+d}{2}\right)}{\Gamma_n\left(\frac{d}{2}\right)} \cdot |\Psi|^{-\frac{T-p}{2}} \cdot |D'_\Omega x' x D_\Omega + I_k|^{-\frac{n}{2}} \cdot \left| I_n + D'_\Psi \left[\hat{\varepsilon}' \hat{\varepsilon} + (\hat{B} - \mathfrak{b})' \Omega^{-1} (\hat{B} - \mathfrak{b}) \right] D_\Psi \right|^{-\frac{T-p+d}{2}}, \quad (\text{A.9})$$

where $D_\Omega D'_\Omega = \Omega$ and $D_\Psi D'_\Psi = \Psi^{-1}$. The last two determinants can be computed as the product of one plus the eigenvalues of $D'_\Omega x' x D_\Omega$ and $D'_\Psi \left[\hat{\varepsilon}' \hat{\varepsilon} + (\hat{B} - \mathfrak{b})' \Omega^{-1} (\hat{B} - \mathfrak{b}) \right] D_\Psi$ respectively, which is numerically stable.

A.3 The ML with dummy observations

It is common in the literature to implement some conjugate priors using dummy observations (e.g. the sum-of-coefficients and the dummy-initial-observation priors). In this case, the ML is given by $p(Y^\oplus)/p(Y^*)$, where $p(\cdot)$ is the function (A.8) or (A.9), Y^* denotes the dummy observations, and $Y^\oplus$ is the extended set of data, consisting of Y and Y^* .

A.4 Proof of the fit-complexity trade-off result

In order to prove (2.5), rewrite (A.8) as

$$p(Y) = \left(\frac{1}{\pi}\right)^{\frac{n(T-p)}{2}} \frac{\Gamma_n\left(\frac{T-p+d}{2}\right)}{\Gamma_n\left(\frac{d}{2}\right)} \cdot \left(\frac{T-p+d+n+1}{d+n+1}\right)^{-\frac{T-p+d}{2}} |\Omega|^{-\frac{n}{2}} \cdot |\Psi|^{-\frac{T-p}{2}} \cdot |x' x + \Omega^{-1}|^{-\frac{n}{2}} \cdot \left| \left(\frac{\Psi + \hat{\varepsilon}' \hat{\varepsilon} + (\hat{B} - \mathfrak{b})' \Omega^{-1} (\hat{B} - \mathfrak{b})}{T-p+d+n+1} \right)^{-1} \frac{\Psi}{d+n+1} \right|^{\frac{T-p+d}{2}}. \quad (\text{A.10})$$

Define $x^t \equiv [x_{p+1}, \dots, x_t]'$ and notice that $x^{t'} x^t$ can be written recursively as $x^{t'} x^t = x^{t-1'} x^{t-1} + x_t x_t'$. The matrix determinant lemma (Harville, 1997) implies that $|x^{t'} x^t + \Omega^{-1}|$ can also be expressed recursively as

$$|x^{t'} x^t + \Omega^{-1}| = |x^{t-1'} x^{t-1} + \Omega^{-1}| \cdot \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t\right). \quad (\text{A.11})$$

The iteration of (A.11), starting from the initial value $|\Omega^{-1}|$, allows to derive

$$|x' x + \Omega^{-1}| = |\Omega^{-1}| \prod_{t=p+1}^T \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t\right).$$

If we substitute this last expression into (A.10), we obtain

$$\begin{aligned}
p(Y) &= \left(\frac{1}{\pi}\right)^{\frac{n(T-p)}{2}} \frac{\Gamma_n\left(\frac{T-p+d}{2}\right)}{\Gamma_n\left(\frac{d}{2}\right)} \cdot \left(\frac{T-p+d+n+1}{d+n+1}\right)^{-\frac{T-p+d}{2}} \\
&\quad |\Psi|^{-\frac{T-p}{2}} \prod_{t=p+1}^T \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t\right)^{-\frac{n}{2}} \\
&\quad \left| \left(\frac{\Psi + \hat{\varepsilon}' \hat{\varepsilon} + (\hat{B} - \mathfrak{b})' \Omega^{-1} (\hat{B} - \mathfrak{b})}{T-p+d+n+1} \right)^{-1} \frac{\Psi}{d+n+1} \right|^{\frac{T-p+d}{2}},
\end{aligned}$$

which, using the properties of the determinants and Kronecker products, can be rewritten as

$$\begin{aligned}
p(Y) &= \left(\frac{1}{\pi}\right)^{\frac{n(T-p)}{2}} \frac{\Gamma_n\left(\frac{T-p+d}{2}\right)}{\Gamma_n\left(\frac{d}{2}\right)} \cdot \left(\frac{T-p+d+n+1}{d+n+1}\right)^{-\frac{T-p+d}{2}} \\
&\quad \prod_{t=p+1}^T \left| \Psi \otimes \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t\right) \right|^{-\frac{1}{2}} \\
&\quad \left| \left(\frac{\Psi + \hat{\varepsilon}' \hat{\varepsilon} + (\hat{B} - \mathfrak{b})' \Omega^{-1} (\hat{B} - \mathfrak{b})}{T-p+d+n+1} \right)^{-1} \frac{\Psi}{d+n+1} \right|^{\frac{T-p+d}{2}}.
\end{aligned}$$

Finally, notice that

$$\begin{aligned}
E_{\Sigma} [\text{var}(y_t | y^{t-1}, \Sigma)] &= E_{\Sigma} \left[X_t \left(\Sigma \otimes (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} \right) X_t' + \Sigma \right] \\
&= E_{\Sigma} \left[\Sigma \otimes \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t \right) \right] \\
&= \frac{\Psi}{d-n-1} \otimes \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t \right),
\end{aligned}$$

where E_{Σ} denotes the expectation operator with respect to Σ . We can now express the ML as

$$\begin{aligned}
p(Y) &= \text{const} \cdot \left| (V_{\varepsilon}^{\text{posterior}})^{-1} V_{\varepsilon}^{\text{prior}} \right|^{\frac{T-p+d}{2}} \cdot \prod_{t=p+1}^T |V_{t|t-1}|^{-\frac{1}{2}} \\
\text{const} &\equiv \left(\frac{1}{\pi} \right)^{\frac{n(T-p)}{2}} \frac{\Gamma_n \left(\frac{T-p+d}{2} \right)}{\Gamma_n \left(\frac{d}{2} \right)} \cdot \frac{(d-n-1)^{\frac{d}{2}}}{(T-p+d-n-1)^{\frac{T-p+d}{2}}} \\
V_{t|t-1} &\equiv E_{\Sigma} [\text{var}(y_t | y^{t-1}, \Sigma)] = \frac{\Psi}{d-n-1} \otimes \left(1 + x_t' (x^{t-1'} x^{t-1} + \Omega^{-1})^{-1} x_t \right) \\
V_{\varepsilon}^{\text{prior}} &\equiv E[\Sigma] = \frac{\Psi}{d-n-1} \\
V_{\varepsilon}^{\text{posterior}} &\equiv E[\Sigma|Y] = \frac{\Psi + \hat{\varepsilon}' \hat{\varepsilon} + (\hat{B} - b)' \Omega^{-1} (\hat{B} - b)}{T-p+d-n-1},
\end{aligned}$$

where $V_{\varepsilon}^{\text{prior}}$ and $V_{\varepsilon}^{\text{posterior}}$ are the prior and posterior means of the residual variance, and their analytical expressions follow from the properties of the Inverse-Wishart distribution.

B The MCMC Algorithm

This appendix presents the details of the MCMC algorithm that we use to simulate the posterior of the coefficients of the BVAR, including the hyperparameters. We use the following Metropolis algorithm:

1. Initialize the hyperparameters γ at their posterior mode, which requires a numerically maximization.
2. Draw a candidate value of the hyperparameters γ^* from a Gaussian proposal distribution, with mean equal to $\gamma^{(j-1)}$ and variance equal to $c \cdot W$, where $\gamma^{(j-1)}$ is the previous draw of γ , W is the inverse Hessian of the negative of the log-posterior of the hyperparameters at the peak, and c is a scaling constant chosen to obtain an acceptance rate of approximately 20 percent.

3. Set

$$\gamma^{(j)} = \begin{cases} \gamma^* & \text{with pr. } \alpha^{(j)} \\ \gamma^{(j-1)} & \text{with pr. } 1 - \alpha^{(j)}, \end{cases}$$

where

$$\alpha^{(j)} = \min \left\{ 1, \frac{p(\gamma^*|y)}{p(\gamma^{(j-1)}|y)} \right\}$$

4. Draw $[\beta^{(j)}, \Sigma^{(j)}]$ from $p(\beta, \Sigma | y, \gamma^{(j)})$, which is the density of the Normal-Inverse-Wishart distribution in (A.3)-(A.4).
5. Increment j to $j+1$ and go to 2.

C Factor Augmented Regression

We consider the following forecasting equation:

$$z_{i,t+h}^h = c_i + \sum_{s=0}^{p_z-1} \alpha_{i,s} z_{i,t-s} + \sum_{k=1}^r \lambda_{ik} f_{k,t} + e_{i,t+h}^h$$

where $z_{i,t+h}^h$ denotes the h -step ahead variable to be forecasted. The predictors $f_{k,t}$, $k = 1, \dots, r$ are common factors extracted from the set of all variables. The lags of the target variable $z_{i,t-s}$ are explicitly used as predictors in order to capture variable specific dynamics. The regression coefficients are allowed to differ across forecast horizons, but the dependence is dropped for notational convenience.

The estimation of the forecasting equation is performed in two steps, as in Stock and Watson (2002a,b). In the first step, the common factors $f_{k,t}$ are estimated by principal components extracted from a large set of 149 predictors. Before extracting the common factors, the data are transformed in order to achieve stationarity and standardized. For details on data definitions and transformations see table 1 and Stock and Watson (2008).

In the second step, the coefficients are estimated by ordinary least squares. Using all the principal components (i.e. by setting r equal to the number of variables 149) would be equivalent to running an OLS regression on all the available regressors. Therefore, as in Stock and Watson (2008), we set $r = 3$ and $p_z = 4$.

References

- HARVILLE, D. (1997): *Matrix Algebra from a Statistician's Perspective*. Springer Verlag.
- STOCK, J. H., AND M. W. WATSON (2002a): "Forecasting Using Principal Components from a Large Number of Predictors," *Journal of the American Statistical Association*, 97, 147–162.
- (2002b): "Macroeconomic Forecasting Using Diffusion Indexes," *Journal of Business and Economics Statistics*, 20, 147–162.
- (2008): "Forecasting in Dynamic Factor Models Subject to Structural Instability," in *The Methodology and Practice of Econometrics, A Festschrift in Honour of Professor David F. Hendry*, ed. by J. Castle, and N. Shephard. Oxford University Press.